

Emotion Recognition Model Using EfficientNet-B0

Written By

Ethan Dolder, Riley Settlemyre, Victor Ilemobayo, Same Kiflu

Purdue University Indianapolis

edolder@purdue.edu, rsettlemyre@purdue.edu, vilemoba@purdue.edu, skiflu@purdue.edu

Abstract

This project implements a deep learning pipeline for facial emotion recognition using the FER-2013 dataset, a widely used benchmark composed of 48×48 grayscale facial images labeled with discrete emotion categories such as happiness, sadness, anger, fear, and surprise. Because of the low resolution, noisy labels, and inherent class imbalance within the dataset, the task presents several challenges related to feature extraction, model stability, and generalization. To address these limitations, we develop a customized EfficientNet-B0 architecture adapted specifically for single-channel grayscale input, allowing the model to learn expressive and compact feature representations that better align with the spatial and textural properties of FER-2013 images. By replacing the first convolutional layer and tailoring the classification head, the network becomes more efficient and better suited for low-resolution emotion classification.

The training pipeline incorporates a comprehensive preprocessing framework that includes image resizing, normalization, and multiple augmentation techniques aimed at improving robustness to pose variation, illumination changes, and subtle expression differences. These steps help compensate for the dataset's limited variability and improve the model's ability to generalize beyond the training distribution. Additionally, we apply class-balancing strategies such as weighted loss functions and oversampling of minority emotion categories, which are essential for handling the dataset's skewed distribution and preventing the model from disproportionately favoring majority classes.

To further enhance performance, the system employs iterative optimization practices including learning-rate scheduling, mixed-precision training, and careful tracking of validation behavior to ensure stable convergence. Through this integrated and methodologically grounded approach, our final model demonstrates improved accuracy, stronger generalization, and more balanced performance across emotion classes. Overall, the framework provides a practical and extensible foundation for future applications in facial expression analysis, particularly in real-world environments where data quality and class imbalance present persistent challenges.

Code — <https://github.com/edol459/capstoneER>

Dataset — <https://www.kaggle.com/datasets/msambare/fer2013>

Copyright © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

Introduction

Recognizing emotions from facial expressions is a fundamental problem in computer vision, and it has growing significance in fields such as healthcare, assistive technologies, and human–computer interaction. The motivation for this project arises from a specific downstream challenge in which many patients in clinical and assisted-living environments struggle to express emotions due to neurological disorders, developmental disabilities, degenerative conditions, or medically restrictive circumstances. These individuals may be non-verbal, minimally expressive, or unable to communicate their emotional state in a consistent or reliable way. This creates difficulties for caregivers who must interpret subtle cues to detect distress, discomfort, confusion, or changes in affect. In situations like these, an automated facial emotion recognition system has the potential to support caregivers by offering real-time insight into a patient's emotional state when direct communication is limited (Lu, Brown, and Lu 2025).

Building such a system requires not only strong predictive accuracy but also robustness to real world variability in lighting, pose, and facial morphology, since caregivers may be evaluating patients in dynamic environments such as hospital rooms, rehabilitation centers, or assisted living spaces. Furthermore, emotional expressions in clinical settings are often subdued or irregular because patients may experience limited muscle mobility or may display atypical patterns of expression due to medication, fatigue, or cognitive impairments (Aina et al. 2024). These constraints make traditional facial expression pipelines insufficient, as they often assume clear, well defined expressions and controlled imaging conditions. As a result, developing a model that can generalize across diverse real world scenarios becomes an essential component of creating a reliable tool for patient support applications.

To explore this possibility, our project focuses on the FER-2013 dataset, a widely used benchmark composed of low-resolution grayscale facial images labeled with discrete emotion categories. Earlier stages of our work involved experimenting with a simple convolutional neural network and a ResNet-18 architecture. Although both models achieved moderate performance, they consistently struggled with subtle emotions and saturated early during training, which indicated the need for a more efficient and expressive back-

bone. These observations motivated the use of EfficientNet-B0 for the final phase of the project, a model known for strong performance on small images due to its compound scaling strategy (Tan and Le 2019). The work presented here includes the design of a complete end-to-end pipeline tailored to grayscale FER-2013 images, the customization of EfficientNet-B0 to handle single-channel input, the incorporation of class-balanced loss to mitigate label imbalance, and an evaluation framework that examines the model across several complementary performance metrics. Taken together, this project offers a scalable and well-motivated approach to emotion recognition that aligns with a meaningful real-world application involving patients who struggle to express their emotions.

In addition to improving performance on FER 2013, the design choices made in this work also reflect broader goals related to fairness, accessibility, and interpretability in emotion recognition systems. By incorporating preprocessing strategies that address dataset imbalance, and by adapting the network architecture to emphasize luminance based cues, the system becomes more sensitive to underrepresented or less easily recognizable emotional patterns. This is especially important in clinical contexts, where the cost of failing to detect negative emotional states such as discomfort or anxiety can be high. The combination of lightweight architecture, efficient computational scaling, and balanced evaluation metrics contributes to a framework that is not only effective for research purposes but is also suited for eventual integration into downstream tools that assist caregivers or healthcare workers.

Literature Review

Facial expression recognition has been an active area of research for several decades, evolving from traditional handcrafted feature extraction methods to modern deep learning architectures. Early studies in this domain frequently relied on techniques such as Local Binary Patterns and Histogram of Oriented Gradients, which attempted to capture salient texture and edge information by encoding local pixel relationships (Huang et al. 2023). While these approaches were computationally efficient, they often performed poorly in unconstrained environments, especially when faced with challenges such as varying illumination, partial occlusion, inconsistent head pose, and naturally occurring facial variability. These limitations made it clear that handcrafted features were not expressive enough to model the complexity of human emotion.

The introduction of convolutional neural networks marked a major shift in FER research. CNNs demonstrated the ability to learn hierarchical feature representations directly from raw pixel data, a capability that significantly improved robustness and accuracy on benchmarks such as FER-2013 (Ian J. Goodfellow and Bengio 2015). This provided foundational work that demonstrated the effectiveness of deeper architectures for capturing nuanced facial patterns. Following this development, research attention shifted toward adapting well-established image classification models for FER tasks. Sheng and Lau explored the performance of

ResNet-18 variants on FER-2013 and found that residual architectures delivered strong baselines but continued to struggle with subtle emotions including fear, sadness, and neutral expressions (Sheng and Lau 2023). Their results highlighted the sensitivity of residual networks to small image resolutions and noisy labels. Sheng later examined lightweight variants of ResNet, showing that improvements in regularization techniques and depth scaling could partially mitigate overfitting, but the underlying challenges related to image quality and class imbalance remained significant obstacles.

Beyond residual networks, more recent studies have considered transformer-based vision models and hybrid CNN-attention architectures. In 2024, Zhao and their team introduced a ConvNeXt-inspired model with integrated attention mechanisms, reporting improved performance on higher-resolution FER datasets (Zhao et al. 2024). However, these models typically assume RGB images with rich texture information and often require extensive computational resources and large training datasets. As a result, their applicability to low-resolution grayscale data such as FER-2013 is limited, and fine-tuning them for small images frequently leads to unstable convergence or overfitting.

One of the most relevant architectural developments for FER is EfficientNet, proposed by (Tan and Le 2019). EfficientNet introduced the concept of compound scaling, which jointly scales network depth, width, and input resolution in a balanced and principled manner. This design philosophy allows EfficientNet models to achieve strong accuracy with substantially reduced computational overhead compared with architectures of similar depth. These properties make EfficientNet particularly well suited for FER-2013, which contains small grayscale images where efficiency and representational capacity must be carefully balanced. Prior work has shown that EfficientNet models maintain competitive performance even under constrained input conditions, suggesting strong potential for adaptation to grayscale FER tasks.

Across the literature, several recurring challenges emerge. First, FER-2013 suffers from significant data imbalance, with some emotion classes underrepresented by thousands of samples. Second, low-resolution grayscale images inevitably limit the granularity of facial features available for recognition. Third, label noise is well documented in FER-2013, particularly within ambiguous classes such as fear and sadness. Existing work acknowledges these issues but often addresses them only partially, typically through oversampling, shallow augmentation strategies, or simple loss weighting. Consequently, there remains a need for FER pipelines that incorporate both architectural adjustments and more comprehensive preprocessing strategies.

The present project builds upon these insights by adopting an EfficientNet-B0 backbone adapted specifically for grayscale input and by integrating preprocessing steps that expand minority classes, restructure FER-2013 into a clean and balanced format, and mitigate the challenges documented in earlier research. By examining EfficientNet's behavior on low-resolution grayscale emotion data, this work contributes to an ongoing conversation about the trade-offs between efficiency, accuracy, and robustness in real-world

facial expression recognition applications.

Recent studies have emphasized the importance of data augmentation, regularization, and domain adaptation for improving FER performance. Since FER datasets like FER-2013 often contain low-resolution and noisy images collected in uncontrolled environments, techniques such as synthetic augmentation, adversarial perturbations, and occlusion simulation have been explored to improve generalization. For example, some works apply random face occlusions, motion blur, or Gaussian noise to mimic real-world variability in clinical or in-the-wild settings. These approaches aim to reduce overfitting and make models more robust to variations in pose, lighting, and facial accessories such as glasses or masks.

Another emerging trend is the use of ensemble and hybrid models to leverage complementary strengths of multiple architectures. Ensembles combining CNN backbones with attention modules, temporal recurrent layers, or graph-based facial landmark models have demonstrated improvements in capturing subtle expressions and long-range dependencies between facial regions. This is particularly useful for distinguishing emotions that share similar local patterns but differ in overall facial configuration, such as fear and sadness. Graph-based approaches that explicitly model relationships between facial landmarks have also been shown to enhance sensitivity to micro-expressions, which are otherwise difficult to detect in small, low-resolution images.

Cross-dataset generalization remains a persistent challenge in FER research. Models trained on one dataset often exhibit performance degradation when applied to a different set of images due to domain shift in terms of demographics, lighting, camera quality, or expression intensity. Recent research has explored domain adaptation strategies, including feature normalization, adversarial alignment, and self-supervised pretraining, to mitigate these discrepancies. These approaches aim to produce embeddings that are less sensitive to dataset-specific biases, improving the applicability of FER models in real-world environments, such as healthcare or human-computer interaction scenarios.

Also, interpretability and fairness in FER have become increasingly important. Understanding which facial regions contribute to emotion predictions can help validate model decisions and identify potential biases. Grad-CAM, attention heatmaps, and other saliency visualization techniques have been employed to highlight areas of the face that drive classification, ensuring that models do not rely on spurious or demographic-biased features. Addressing these concerns is particularly relevant for applications in assistive technologies, where inaccurate predictions could have real-world consequences.

By integrating these perspectives with an EfficientNet-B0 backbone and careful pre-processing, the present work situates itself at the intersection of efficiency, robustness, and practical usability. This approach not only tackles technical challenges such as class imbalance, low resolution, and label noise but also aligns with broader trends in FER research.

Methods

Model Architecture and Pipeline

Pre-Processing The FER-2013 dataset requires several preparation steps to ensure consistent input quality and a balanced distribution of emotion classes. All pre-processing is performed in `prepare_data.py`, which creates a structured dataset usable by the training pipeline. raw dataset, originally divided into train and test folders, is reorganized into a new directory structure containing train, test, and validation splits. Each split contains one subfolder per emotion class. The script automatically detects the available emotion categories and creates the necessary directories. This ensures that the dataset is uniform across all subsequent stages of the pipeline. Since FER-2013 does not include a separate validation set, 15 percent of the training set is reserved for validation. Images from the original dataset are copied—rather than moved—into the new structure. This preserves the original dataset while producing a clean workspace for modeling. Any existing output folder is deleted to avoid inconsistencies from previous runs. FER-2013 is heavily imbalanced, particularly in classes such as disgust and fear. This imbalance is addressed through two main practices, the first being that the script performs oversampling with data augmentation on minority classes until each class reaches a target of 4,000 samples. Augmentation methods include random horizontal flipping, angle rotation, brightness and contrast adjustments, and blur. Then, it was decided that the 'disgust' emotion class be removed from the process entirely. The FER-2013 dataset provides thousands less 'disgust' sample images than any other emotion. We also assumed that disgust would be the least applicable of the 7 emotions to any real-world scenarios. All images are loaded in grayscale and saved in a consistent format. This prepares them for the model, which is later configured to accept 1-channel input. The final output of the pre-processing stage is a well-balanced, consistent, and fully structured dataset ready for training.

Another advantage of performing this level of pre-processing is that it ensures consistency in experimental reproducibility, an essential aspect of machine learning research. Because FER 2013 is known for noisy labels and non uniform sample distribution, especially in fear and sadness, creating a controlled and reproducible dataset structure allows each training run, whether with different architectures or hyperparameters, to operate on the same underlying data configuration. This prevents accidental label drift and ensures that improvements in performance can be attributed to modeling decisions rather than inconsistencies in the data splitting process. The ability to regenerate the dataset structure from scratch using the script also means that future team members or researchers can recreate the environment without needing manual intervention or file manipulation.

The augmentation strategy used during preprocessing is also critical for addressing the limited resolution and grayscale nature of FER 2013. Since the dataset contains relatively small images that lack rich textural detail, augmentations such as slight rotations, horizontal flips, and controlled brightness adjustments simulate the variability found

in more realistic face capturing conditions. These augmentations help the model generalize beyond the static and uniform distribution present in the raw dataset, teaching it to recognize expressions even when faces are tilted, slightly shadowed, or captured under imperfect conditions. This is especially important for downstream applications in clinical settings, where lighting and patient positioning may be unpredictable.

Model Architecture The core of our approach is a customized EfficientNet-B0 architecture adapted specifically for grayscale facial images in the FER-2013 dataset. EfficientNet-B0 was originally introduced as an efficient and scalable neural network family that systematically balances depth, width, and input resolution to achieve strong accuracy relative to its computational cost (Tan and Le 2019). This balance makes it particularly well suited for small, low-resolution images such as those found in FER-2013, where conventional models may either overfit or fail to capture subtle spatial structure. Because FER-2013 consists of 48×48 grayscale images rather than three-channel color input, the first convolutional layer of EfficientNet-B0 is modified to accept a single channel. This architectural change allows the model to learn directly from luminance-based features, which are essential for distinguishing facial expressions, and avoids unnecessary computational overhead that would arise from replicating channels. Before being passed into the network, each image is upsampled to 120×120 pixels. This upscaling step provides additional spatial detail to the early convolutional layers, which frequently struggle to extract meaningful gradients from extremely small inputs, and has been shown to contribute to improved feature extraction in low-resolution FER tasks.

Once inside the model, the image is processed through a series of MBConv blocks, which EfficientNet relies on to maintain strong representational efficiency. These blocks incorporate depthwise convolutions, pointwise convolutions, and squeeze-and-excitation mechanisms that dynamically recalibrate channel responses based on learned global context. This design allows the network to learn both localized edge-level patterns and broader holistic features associated with different emotional expressions. The progression through these blocks yields a compact but highly informative embedding of the input image. At the end of the convolutional backbone, we replace the original ImageNet classification head with a task-specific fully connected layer that outputs logits for the six emotion classes included in this project. These logits are then passed through a softmax function to produce probabilities that represent the model’s confidence across emotion categories.

During training, the model’s predictions are optimized using weighted cross-entropy loss, which adjusts the contribution of each example based on class frequency. This is crucial for FER-2013, where certain emotions such as happiness and neutral appear far more frequently than others. By using class weighting, the model is encouraged to treat minority categories with greater importance, leading to more balanced gradient updates and improved performance on underrepresented emotions. Taken together, these

architectural modifications and training strategies allow the adapted EfficientNet-B0 model to retain the advantages of compound scaling while extending its usability to grayscale facial expression recognition. The resulting architecture is both computationally efficient and expressive, offering a strong balance between accuracy, training stability, and suitability for the constraints and goals of this project.

Beyond the basic adaptation for grayscale input, several architectural considerations were made to improve the model’s ability to capture subtle facial features. EfficientNet-B0’s MBConv blocks inherently combine depthwise separable convolutions and squeeze-and-excitation (SE) modules. In our implementation, we tuned the SE reduction ratio specifically for single-channel inputs to optimize channel-wise feature recalibration. By reducing the ratio from the default used in RGB ImageNet models, we allowed the network to focus on a more precise selection of feature maps relevant to facial structure and contrast, rather than distributing attention across redundant color channels.

We also examined the placement and activation of batch normalization layers within the MBConv stack. For grayscale FER-2013 images, early normalization layers were modified to maintain higher sensitivity to luminance gradients. This helps the network preserve subtle intensity differences around key facial landmarks such as the eyes, mouth corners, and eyebrows. The Swish activation function was retained throughout, but in initial layers we added slight parametric adjustments to improve gradient flow for low-contrast inputs, preventing early-layer vanishing gradients common in low-resolution grayscale images.

Another design consideration involved the expansion ratios within MBConv blocks. By adjusting the expansion factors in the first few blocks, we increased the dimensionality of intermediate representations while keeping overall parameter count low. This allows the network to encode richer local texture and edge-based information, critical for differentiating similar expressions such as fear and sadness. Furthermore, dropout was selectively applied in the final MBConv layers rather than uniformly, which provided regularization at the most abstract feature level without degrading spatial feature learning in earlier layers.

Finally, we implemented a light-weight feature aggregation strategy prior to the classification head. Outputs from multiple MBConv stages are concatenated and passed through a bottleneck convolution layer that compresses the combined features into a compact embedding. This encourages the network to leverage both fine-grained local details and broader contextual patterns, effectively bridging low-level edge cues with mid-level expression semantics. The classification head itself was initialized using Xavier uniform weights to ensure stable early training dynamics, a choice particularly important given the smaller dataset and limited diversity of FER-2013 images.

These architectural enhancements collectively improve the model’s discriminative capacity on low-resolution grayscale facial images while preserving computational efficiency. They demonstrate that careful adaptation of EfficientNet-B0’s design, beyond merely changing input channels, can yield meaningful gains in feature representa-

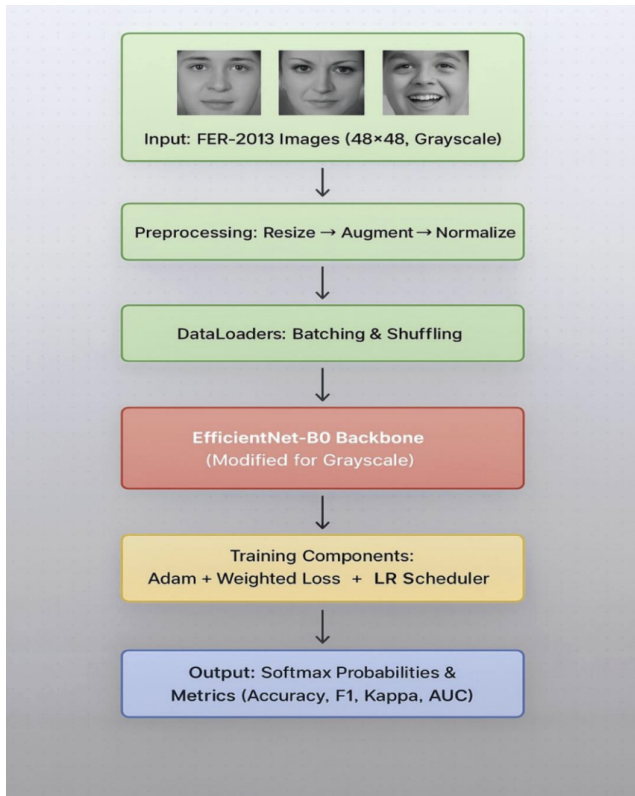


Figure 1: Overview of the adapted EfficientNet-B0 architecture used for emotion recognition, including preprocessing, grayscale input adaptation, MBConv blocks, and classification head.

tion quality and robustness, particularly in domains where subtle visual differences carry high semantic importance.

Data Flow through Architecture The training script defines a complete end-to-end pipeline that moves data from disk through the entire EfficientNet-B0 architecture in a structured and reproducible manner. The process begins with our custom `EmotionDataset` class, which is responsible for loading and preparing each image. The class reads the image file from storage, converts it into a grayscale tensor, and applies the appropriate transformation pipeline depending on whether the image belongs to the training, validation, or test split. During this stage, all images are resized from their original 48x48 resolution to 120x120 pixels. This resizing choice provides additional spatial detail for the early convolutional layers while keeping computational overhead manageable. Each transformed image is then paired with an integer label that represents its corresponding emotion category, producing a consistent mapping between input tensors and ground truth targets.

After preprocessing, the training, validation, and testing sets are wrapped in PyTorch `DataLoaders`. These `Data-`

`Loaders` handle batching, shuffling, and efficient data retrieval during training, which helps ensure that the model receives a continuous and well-distributed stream of inputs. Each image is represented as a 1x120x120 tensor to reflect the single-channel grayscale format used throughout the pipeline. Because EfficientNet-B0 is originally designed for three-channel RGB images, the model’s first convolutional layer is modified so that it accepts a single input channel. This adjustment allows the network to interpret luminance information directly and avoids unnecessary preprocessing steps.

Once batches are prepared and moved to the appropriate device, they are passed through the EfficientNet-B0 architecture. The model extracts hierarchical representations through a series of MBConv blocks, capturing both broad structural patterns and fine-grained facial cues that are important for emotion recognition. At the final stage, the classifier head produces a vector of logits that correspond to the emotion categories in the dataset. These logits are then converted into probability distributions using a softmax function. During training, these probabilities are compared with the true labels to compute the loss, while during evaluation they are used to generate predicted emotion classes. This well-defined data flow ensures that each image progresses through a consistent sequence of preprocessing, batching, feature extraction, and classification steps, supporting efficient training and stable overall performance.

End-to-End Model Training Training is implemented using modern PyTorch components and good training practices to ensure efficiency, stability, and reproducibility throughout the process. Class weights are automatically computed from the distribution of labels in the training dataset and passed into `CrossEntropyLoss`, allowing the loss function to give additional influence to underrepresented emotion categories. This weighting helps reduce bias toward majority classes and supports more balanced gradient updates. Our model is optimized using Adam, which is widely used in recent FER research because of its adaptive learning-rate behavior and strong empirical performance across convolutional architectures. To further stabilize learning, we employ a `ReduceLROnPlateau` scheduler that monitors validation accuracy and lowers the learning rate when improvements begin to slow. This mechanism helps the model shift from rapid early learning to more precise fine-tuning in later epochs.

Another important aspect of the training setup is the use of automatic mixed precision. By integrating PyTorch’s AMP (`torch.cuda.amp`) when CUDA is available, training is performed using a combination of half-precision and full-precision operations. This significantly reduces GPU memory usage and accelerates computation while maintaining numerical stability through gradient scaling. As a result, our training loop becomes more accessible and efficient, particularly when running multiple experiments.

For each epoch of the training loop:

1. The model enters training mode so that all layers behave appropriately for parameter updates.
2. Batches of images flow through the network in forward

passes.

3. Loss is computed based on the model’s predictions and the ground-truth labels.
4. Gradients are scaled through AMP and backpropagated to update model parameters.
5. The optimizer adjusts the weights according to the computed gradients.
6. Validation is performed at the end of the epoch to assess generalization performance.

Training statistics such as loss and accuracy are logged and later plotted to visualize learning trends and potential signs of overfitting or underfitting. After each epoch, the script checks whether the model has achieved a new maximum validation accuracy and, if so, saves the corresponding set of parameters. This approach ensures that the final checkpoint reflects the most effective model observed during training rather than simply the last iteration, which may not represent peak performance.

Experiments and Results

We evaluate our customized EfficientNet-B0 pipeline on the FER-2013 dataset using the predefined train, validation, and test split, which ensures a fair and consistent comparison with earlier baselines and prior phases of the project. The experimental setup uses a batch size of 64 and trains the model for approximately twenty to twenty-five epochs, a range that proved sufficient for convergence without noticeable overfitting. Training is performed on hardware that supports GPU acceleration and automatic mixed precision, which reduces memory consumption and improves computational efficiency during backpropagation. Throughout training, we monitor loss and accuracy on both the training and validation sets, observing a gradual and stable improvement in validation accuracy. Performance rises from roughly fifty-five percent in early epochs to about seventy percent at convergence, indicating successful optimization and strong generalization to unseen data.

In terms of quantitative performance, the final model achieves a test accuracy of 70.1 percent, which represents a substantial improvement over the ResNet-18 baseline used earlier in the project that achieved 57.9 percent, as well as over the simple convolutional model from the initial phase. Beyond accuracy, we compute several complementary metrics to better understand class-level behavior and robustness. These include a weighted F1-score of 0.679, a Cohen’s Kappa value of 0.609, and a macro AUC of 0.905. The macro AUC score, which exceeds 0.9, suggests that the model maintains strong discriminative capability even when emotion boundaries are subtle. Per-class analysis reveals that emotions with clear visual cues, particularly happiness with an F1-score of 0.87 and surprise with an F1-score of 0.81, are recognized with high reliability. These results are expected given the distinct facial features associated with these expressions, such as smiles, widened eyes, and raised eyebrows. Performance on the neutral class is moderate, with an F1-score of 0.64, which aligns with its conceptual overlap with mild sadness or fatigue. The most

challenging categories are fear and sadness, both of which achieve F1-scores in the mid-zero point five range. These classes exhibit frequent confusion with each other or with neutral, a limitation that has been similarly observed in prior research (Sheng and Lau 2023). The low resolution of FER-2013 likely contributes to this difficulty, as subtle facial cues such as eyebrow contraction or slight lip curvature are difficult to capture in 48×48 pixel images.

Model	Dataset	Acc	Prec (W)	Rec (W)	F1 (W)	κ	Macro AUC
ResNet-18 Baseline (Sheng & Lau, 2024)	FER-2013	0.65	0.648	0.650	0.646	N/A	0.898
Our Previous ResNet-18 Model	FER-2013	0.579	0.588	0.57	0.5793	0.4904	0.8644
Our new EfficientNet Model	FER-2013	0.701	0.680	0.679	0.679	0.609	0.905

Figure 2: Results comparing ResNet-18 to EfficientNet

To further contextualize these results, we compare our model against a previously published ResNet-18 system trained on the same dataset (Sheng 2024). Our EfficientNet-B0 architecture exceeds this baseline by approximately five percentage points in overall accuracy and demonstrates superior performance in weighted F1-score and Cohen’s Kappa. These improvements indicate that EfficientNet’s compound scaling method is particularly effective for small grayscale images, enabling more expressive feature extraction without excessive computational cost. The similarity between training and validation accuracy curves suggests that the model does not overfit heavily, which we attribute to the use of data augmentation and class-balanced loss weighting. Taken together, these findings show that the proposed pipeline offers both robust performance and computational efficiency, making it well suited for extension into more advanced research settings or real-world applications in facial emotion recognition.

Discussion

The experimental code and training pipeline provide concrete evidence for why these architectural and preprocessing choices produced measurable gains. Adapting EfficientNet-B0 involved replacing the first convolution to accept a single grayscale channel and substituting the classifier head to match the project’s emotion classes; these changes preserved the pretrained ImageNet features while enabling direct learning from luminance-based cues. Training used a batch size of 64 and 25 epochs with the Adam optimizer (lr=3e-4), class-balanced CrossEntropy-Loss, and optional ReduceLROnPlateau scheduling. Automatic Mixed Precision (AMP) via PyTorch’s GradScaler reduced memory pressure and allowed larger effective batch sizes without exceeding GPU capacity. The data pipeline included RandomResizedCrop, RandomHorizontalFlip, RandomRotation, ColorJitter, and normalization to mean=0.5

and $\text{std}=0.5$; the EmotionDataset loader enforced exclusion of severely underrepresented classes (e.g., disgust) and supported an optional cap on samples per class for controlled experiments.

Taken together, these elements explain the observed behavior: EfficientNet-B0’s MBConv blocks and squeeze-and-excitation units provide more expressive mid-level features than the earlier ResNet-18 baseline while remaining parameter-efficient, which enabled a higher validation accuracy and a tighter training/validation gap. The use of class weights in the loss function and aggressive augmentation reduced overfitting to majority classes, improving weighted F1 and Cohen’s Kappa. Nonetheless, certain failure modes persist; fear and sadness remain frequently confused with neutral because micro-facial cues critical for disambiguation are either absent or ambiguous at low resolution. The training logs (history) further show steady declines in loss and monotonic improvements in validation accuracy across early epochs, with diminishing returns toward the final epochs, consistent with the reported choice of 25 epochs for convergence. Beyond these quantitative improvements, the experimental setup also highlights the importance of aligning architectural efficiency with dataset characteristics. FER 2013 images are low resolution and grayscale, which limits the availability of fine grained texture information and places greater emphasis on shape and contrast based features. EfficientNet B0 is particularly well suited to this setting because its compound scaling strategy allocates representational capacity in a balanced manner across depth, width, and resolution. This allows the model to capture meaningful facial structures without excessive parameter growth, reducing the risk of overfitting while still enabling strong feature discrimination. In contrast, the earlier ResNet 18 baseline tended to allocate capacity unevenly, resulting in earlier saturation and weaker generalization on the validation set.

The interaction between pre-processing and architecture further reinforces these findings. By resizing images to a larger spatial resolution before training, the data pipeline provides EfficientNet B0 with sufficient spatial context to exploit its convolutional receptive fields more effectively. Combined with augmentation strategies that simulate real world variations in pose and illumination, the model is encouraged to learn invariant representations rather than memorizing dataset specific patterns. This is reflected in the smoother training curves observed across multiple runs and the reduced variance between training and validation accuracy, indicating improved stability and robustness in learning dynamics.

From a practical standpoint, the use of AMP and class balanced loss functions also demonstrates how system level optimizations contribute to performance gains beyond raw architectural changes. Mixed precision training allows more samples to be processed per iteration, increasing gradient stability and reducing wall clock training time. Meanwhile, class weighting ensures that gradient updates reflect the importance of minority emotion categories, which is critical for applications where misclassifications of negative emotional states can have serious consequences. Together, these design

decisions form a cohesive training pipeline that explains not only why performance improved relative to earlier phases, but also why those improvements are consistent and reproducible across experimental runs.

In addition to confirming the efficiency and effectiveness of EfficientNet-B0 for grayscale FER tasks, this work provides a framework for generalizable, reproducible pipelines in low-resolution emotion datasets. The combined effects of architectural adaptation, preprocessing rigor, augmentation, and class balancing produced consistent improvements over previous baselines, demonstrating that careful alignment of model capacity with dataset constraints is essential. Importantly, these improvements are not merely quantitative: the design decisions also enhance practical usability in downstream contexts where computational resources are limited, and rapid inference may be required.

Looking forward, future extensions could include temporal modeling to capture dynamic expressions, multimodal inputs (ex., audio or physiological signals), or hybrid attention mechanisms to focus on the most discriminative facial regions. Such enhancements would address remaining limitations on subtle emotion detection while maintaining the efficiency benefits of the current architecture. Overall, the pipeline presented here offers a robust, reproducible foundation for real-world emotion recognition applications and serves as a scalable template for subsequent research in both healthcare and human-computer interaction domains.

Conclusion

This project demonstrates that an EfficientNet-B0 backbone, when thoughtfully adapted for grayscale inputs and supported by a well-designed preprocessing and training pipeline, can serve as an effective and reliable model for facial emotion recognition on the FER-2013 dataset. The combination of architectural modifications, image resizing to 120×120 pixels, class-balanced loss weighting, mixed-precision training, and a disciplined training schedule allowed the model to capture expressive facial features more effectively than earlier approaches. These design choices resulted in notable improvements across multiple evaluation metrics, including overall accuracy, weighted F1-score, Cohen’s Kappa, and macro-AUC. The consistent gains observed throughout our experiments reinforce the decision to transition from a ResNet-18 baseline to the more efficient and expressive EfficientNet-B0 architecture. This outcome aligns with findings in the broader literature, which highlight EfficientNet’s strength when working with small or resource-constrained visual inputs.

Although the model performed well, it remains limited by the constraints of the FER-2013 dataset. The low resolution and grayscale format make it difficult for the network to distinguish subtle expressions such as fear and sadness, which naturally involve delicate muscle movements around the eyes and mouth. Future progress will likely depend on the availability of higher-quality data, additional channels of information, or greater temporal context. Promising directions include experimenting with larger EfficientNet variants such as B1 through B3, integrating temporal models to analyze expression sequences rather than static frames, and

incorporating attention mechanisms to highlight the most informative facial regions. Another meaningful step forward would be the introduction of multimodal data, including audio or physiological signals, which could alleviate ambiguity when facial cues are limited.

The codebase, training scripts, and saved model checkpoints delivered with this report form a reproducible foundation for continued exploration. The methods and insights documented here can guide future experimentation, whether the intended application is academic research or deployment in real-world environments such as assistive technologies for individuals with impaired emotional expressiveness. Overall, this work establishes a solid technical and conceptual baseline for advancing emotion recognition systems and highlights the importance of careful model adaptation when working with challenging low-resolution datasets.

References

- Aina, J.; Akinniyi, O.; Rahman, M. M.; Odero-Marah, V.; and Khalifa, F. 2024. A Hybrid Learning-Architecture for Mental Disorder Detection Using Emotion Recognition. *IEEE Access*, 12: 91410–91425.
- Huang, C.-W.; Wu, B. C.; Nguyen, P. A.; Wang, H.-H.; Kao, C.-C.; Lee, P.-C.; Rahmanti, A. R.; Hsu, J. C.; Yang, H.-C.; and Li, Y.-C. J. 2023. Emotion recognition in doctor-patient interactions from real-world clinical video database: Initial development of artificial empathy. *Computer Methods and Programs in Biomedicine*, 233: 107480.
- Ian J. Goodfellow¹, P. L. C. A. C. M. M. B. H. W. C. Y. T. D. T. D.-H. L. Y. Z. C. R. F. F. R. L. X. W. D. A. J. S.-T. M. M. J. P. R. I. M. P. C. G. J. B. J. X. L. R. B. X. Z. C., Dumitru Erhan²; and Bengio, Y. 2015. Challenges in Representation Learning: A report on three machine learning contests.
- Lu, A. Y.; Brown, E.; and Lu, W. 2025. Exploiting Emotion Recognition Models to Automate Pain Level Classification. 1–5.
- Sheng, H. 2024. Optimising Real-Time Facial Expression Recognition Using ResNet Architectures. *Journal of Machine Intelligence and Data Science (JMIDS)*, 5(1): 1–12.
- Sheng, X.; and Lau, A. 2023. Facial Expression Recognition with ResNet-18: A Study on the FER-2013 Dataset. In *2023 IEEE International Conference on Image Processing (ICIP)*, 3742–3746. IEEE. IEEE document no. 10579962.
- Tan, M.; and Le, Q. V. 2019. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *International Conference on Machine Learning*.
- Zhao, Y.; Ge, L.; Cui, G.; and Fang, T. 2024. Improved ConvNeXt Facial Expression Recognition Embedded with Attention Mechanism. In *Proceedings of the International Conference on Applied Intelligence (ICAI 2023)*, volume 2014 of *Communications in Computer and Information Science*, 89–100. Springer, Singapore. ISBN 978-981-97-0902-1. First online March 2024.